Angular Momentum Representation For Free particle

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Abstract

We analyze two indistinguishable fermions in a 1D infinite well with a $g\delta(x_1 - x_2)$ contact potential. We demonstrate that the required spatial antisymmetry forces the wavefunction to vanish at particle coincidence ($\Psi(x, x) = 0$), rendering the interaction inert. The system is then solved as two non-interacting particles. We derive the position-space eigenfunctions, the momentum-space wavefunctions via Fourier transform, and the two-particle propagator, which takes the form of a Slater determinant.

We establish a general theorem for single-particle operator expectation values, $\langle \hat{A}_1 \rangle = \frac{1}{2}(\langle \hat{a} \rangle_{n_1} + \langle \hat{a} \rangle_{n_2})$, using it to calculate uncertainties and verify the Heisenberg principle. Finally, we construct generalized, n-dependent ladder operators, showing the system admits a deformed Heisenberg algebra consistent with its non-linear spectrum.

Keywords: Identical Particles, Infinite Potential Well, Contact Interaction

1 Problem Setting

The purpose of this project is to construct a complete and self-contained formulation of the angular-momentum representation for a nonrelativistic free particle. The central objective is to solve the time-dependent Schrödinger equation (TDSE)

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \qquad \hat{H} = \frac{\hat{\mathbf{p}}^2}{2m},$$

in a basis that diagonalizes simultaneously the free Hamiltonian $\hat{H}$ and the angular-momentum operators $\hat{L}^2$ and $\hat{L}_z$.

The motivation for this construction is threefold:

1. To reveal the role of rotational symmetry in quantum mechanics through the generators of the group SO(3).
2. To establish a clear connection between spatial symmetries and conservation laws.
3. To demonstrate how angular-momentum eigenstates provide a natural separation of variables for free-particle dynamics in spherical coordinates.

1.1 Physical Scenario

We consider a single spinless particle of mass m moving freely in three-dimensional space. No external forces act on the system, so the Hamiltonian contains only kinetic energy:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m}.$$

Because the dynamics is invariant under translations and rotations, the corresponding generators $\hat{\mathbf{p}}$ and $\hat{\mathbf{L}}$ play a central role. In particular:

$$[\hat{H}, \hat{\mathbf{L}}] = 0,$$

which implies that $\hat{H}$, $\hat{L}^2$ and $\hat{L}_z$ admit simultaneous eigenstates.

1.2 Goal of the Construction

We aim to:

- Construct the operators $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ from the canonical commutation relations.
- Express $\hat{\mathbf{L}}$ in both Cartesian and spherical coordinates.
- Establish the commutation relations that define the Lie algebra of SO(3):

$$[\hat{L}_i, \hat{L}_j] = i\hbar\, \epsilon_{ijk} \hat{L}_k.$$

- Solve the eigenvalue problems for $\hat{L}_z$ and $\hat{L}^2$, obtaining the spherical harmonics $Y_{\ell m}(\theta, \phi)$.
- Expand the free-particle wavefunction in the basis $|k, \ell, m\rangle$.
- Solve the TDSE in this basis and reconstruct the wavefunction in position, momentum, and energy representations.

1.3 Mathematical Background Assumed

The development assumes familiarity with:

- Hilbert-space formalism: kets, operators, inner products.
- Canonical commutators $[\hat{x}_i, \hat{p}_j] = i\hbar\, \delta_{ij}$.
- Fourier transforms and the relation between position and momentum representations.

- Basic knowledge of spherical coordinates and special functions (Legendre polynomials).

1.4 Expected Outcome

The final result of this construction is the full decomposition

$$|\psi(t)\rangle = \sum_{\ell,m} \int_0^\infty dk\, k^2\, \psi_{k\ell m}(0)\, e^{-i\hbar k^2 t/(2m)}\, |k, \ell, m\rangle\,,$$

along with the corresponding integral transforms that yield the wavefunction in position-, momentum-, and energy-space representations. This framework encapsulates the full rotational structure of the free particle and sets the stage for generalizations to interacting systems, spin, and relativistic dynamics.

2 Starting Point: Known Results

In this section, we summarize the fundamental elements of the quantum mechanical formalism for a free particle of mass m. These results are assumed to be known and serve as the foundation for constructing the angular momentum representation. Proofs are omitted; only the necessary definitions and operational relations are presented.

2.1 Free Particle Solution in Different Representations

The free particle Hamiltonian is given by $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m}$. The wavefunctions in the various representations evolve as follows:

- **Position Representation ($\psi(\mathbf{r}, t)$):** Satisfies the time-dependent Schrödinger equation:
$$i\hbar \frac{\partial}{\partial t}\psi(\mathbf{r}, t) = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}, t). \tag{1}$$
- **Momentum Representation ($\psi(\mathbf{p}, t)$):** Since $\hat{\mathbf{p}}$ commutes with $\hat{H}$ for the free particle, the evolution is a multiplicative phase:
$$\psi(\mathbf{p}, t) = \psi(\mathbf{p}, 0) \exp\left(-\frac{i}{\hbar}\frac{p^2}{2m}t\right). \tag{2}$$
- **Energy Representation ($\psi(E, t)$):** For an energy eigenstate with $E = p^2/2m$:
$$\psi(E, t) = \psi(E, 0) \exp\left(-\frac{i}{\hbar}Et\right). \tag{3}$$

2.2 Integral Transformations Between Representations

The connection between different bases is established through integral transformations defined by the inner products (kernels) of the basis states.

2.2.1 Fourier Transform ($\mathbf{r} \leftrightarrow \mathbf{p}$)

The wavefunctions are related by:

$$\psi(\mathbf{r}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3p\, e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi(\mathbf{p}, t), \tag{4}$$

$$\psi(\mathbf{p}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3r\, e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \psi(\mathbf{r}, t). \tag{5}$$

2.2.2 Momentum-Energy Transformation ($\mathbf{p} \leftrightarrow E$)

Using the dispersion relation $E = p^2/2m$ and the properties of the deformed Dirac delta:

$$\delta\left(E - \frac{p^2}{2m}\right) = \frac{m}{p}\delta(p - \sqrt{2mE}). \tag{6}$$

This allows expressing integrals over momentum as integrals over energy, facilitating the transition from $\psi(\mathbf{p})$ to spectral energy densities.

2.3 Eigenvalue Equations and Basic Operators

We define the fundamental operators and their eigenstates, which form complete and orthonormal bases.

- **Position Operator $\hat{\mathbf{r}}$:**

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle, \quad \int d^3r |\mathbf{r}\rangle\langle\mathbf{r}| = \hat{\mathbb{I}}. \tag{7}$$

- **Momentum Operator $\hat{\mathbf{p}}$:**

$$\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle, \quad \int d^3p |\mathbf{p}\rangle\langle\mathbf{p}| = \hat{\mathbb{I}}. \tag{8}$$

  In the position basis, the momentum operator acts as the generator of translations: $\hat{\mathbf{p}} = -i\hbar\nabla_{\mathbf{r}}$.
- **Free Hamiltonian $\hat{H}$:** Its eigenstates $|E\rangle$ coincide (with degeneracy) with the momentum states:

$$\hat{H}|\mathbf{p}\rangle = \frac{p^2}{2m}|\mathbf{p}\rangle. \tag{9}$$

- **Inner Products (Transformation Kernels):**

$$\langle\mathbf{r}|\mathbf{p}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}, \quad \langle\mathbf{p}|E\rangle \propto \delta(E - p^2/2m). \tag{10}$$

2.4 Symmetry Operators and Ehrenfest's Theorem

The following unitary operators generate the continuous transformations relevant to the system.

2.4.1 Translation and Evolution Operators

- **Spatial Translation:** Generated by the momentum operator $\hat{\mathbf{p}}$. For a finite vector $\mathbf{a}$ and an infinitesimal one $d\mathbf{a}$:

$$\hat{T}(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right), \quad \hat{T}(d\mathbf{a}) \approx \hat{\mathbb{I}} - \frac{i}{\hbar}d\mathbf{a}\cdot\hat{\mathbf{p}}. \tag{11}$$

- **Time Evolution:** Generated by the Hamiltonian $\hat{H}$.

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\hat{H}t\right), \quad \hat{U}(dt) \approx \hat{\mathbb{I}} - \frac{i}{\hbar}\hat{H}dt. \tag{12}$$

- **Momentum Space Translation:** Analogous to spatial translation, the position operator $\hat{\mathbf{r}}$ generates displacements in momentum space (boosts). For a momentum displacement $\mathbf{q}$:

$$\hat{T}_{\mathbf{p}}(\mathbf{q}) = \exp\left(\frac{i}{\hbar}\mathbf{q}\cdot\hat{\mathbf{r}}\right). \tag{13}$$

  The action of these operators defines the change of basis between the Hilbert spaces associated with the observables.

2.4.2 Correspondence Principle and Ehrenfest's Theorem

The connection to classical mechanics is formally established via Ehrenfest's Theorem, which governs the time evolution of expectation values. For an operator $\hat{A}$ with no explicit time dependence:

$$\frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle. \tag{14}$$

Applied to the free particle ($\hat{H} = \hat{p}^2/2m$):

- **Position:** $\frac{d}{dt}\langle\hat{\mathbf{r}}\rangle = \frac{1}{m}\langle\hat{\mathbf{p}}\rangle$ (Velocity-Momentum relation).
- **Momentum:** $\frac{d}{dt}\langle\hat{\mathbf{p}}\rangle = 0$ (Conservation of linear momentum due to translational invariance).

These results ensure that wave packets follow classical trajectories in the macroscopic limit, satisfying the correspondence principle.

3 Definition of the Angular Momentum Operator

In classical mechanics, the angular momentum of a particle with position vector $\mathbf{r}$ and linear momentum $\mathbf{p}$ is defined as the vector product $\mathbf{L}_{cl} = \mathbf{r}\times\mathbf{p}$. To transition to quantum mechanics, we apply the correspondence principle, promoting the dynamical variables to linear operators acting on the Hilbert space: $\mathbf{r}\to\hat{\mathbf{r}}$ and $\mathbf{p}\to\hat{\mathbf{p}}$.

However, since $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are non-commuting operators, the definition of the vector product must be handled with algebraic rigor to ensure the resulting operator is Hermitian (measurable) and unambiguous.

3.1 The Cross Product of Vector Operators

Let $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ be two vector operators in $\mathbb{R}^3$. The i-th component of their cross product, $(\hat{\mathbf{A}} \times \hat{\mathbf{B}})_i$, is defined using the Levi-Civita completely antisymmetric tensor ϵ_{ijk}, where indices i, j, k run over $\{1, 2, 3\}$ (identifying $1 \to x, 2 \to y, 3 \to z$):

$$(\hat{\mathbf{A}} \times \hat{\mathbf{B}})_i = \sum_{j,k} \epsilon_{ijk} \hat{A}_j \hat{B}_k. \tag{15}$$

Applying this definition to the position and momentum operators, we define the orbital angular momentum operator $\hat{\mathbf{L}}$ as:

$$\hat{\mathbf{L}} \equiv \hat{\mathbf{r}} \times \hat{\mathbf{p}}. \tag{16}$$

In component form, this definition implies:

$$\hat{L}_i = \epsilon_{ijk} \hat{x}_j \hat{p}_k, \tag{17}$$

where we have adopted the Einstein summation convention for repeated indices j and k.

3.2 Non-Commutativity and Hermiticity

A fundamental postulate of quantum mechanics is that physical observables are represented by Hermitian operators ($\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger$). In general, the product of two Hermitian operators $\hat{A}\hat{B}$ is not Hermitian unless $[\hat{A}, \hat{B}] = 0$. Since position and momentum do not commute ($[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk}$), we must verify that the definition in Eq. (17) yields a Hermitian operator without the need for manual symmetrization.

We take the adjoint of the i-th component:

$$\hat{L}_i^\dagger = (\epsilon_{ijk} \hat{x}_j \hat{p}_k)^\dagger. \tag{18}$$

Using the property $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$ and the fact that $\hat{\mathbf{x}}$ and $\hat{\mathbf{p}}$ are Hermitian ($\hat{x}_j^\dagger = \hat{x}_j$, $\hat{p}_k^\dagger = \hat{p}_k$) and ϵ_{ijk} is real:

$$\hat{L}_i^\dagger = \epsilon_{ijk} \hat{p}_k^\dagger \hat{x}_j^\dagger = \epsilon_{ijk} \hat{p}_k \hat{x}_j. \tag{19}$$

Notice the order of operators has reversed. To restore the original order, we use the canonical commutation relation:

$$[\hat{x}_j, \hat{p}_k] = i\hbar\delta_{jk} \implies \hat{p}_k \hat{x}_j = \hat{x}_j \hat{p}_k - i\hbar\delta_{jk}. \tag{20}$$

Substituting this back into the expression for $\hat{L}_i^\dagger$:

$$\hat{L}_i^\dagger = \epsilon_{ijk}(\hat{x}_j \hat{p}_k - i\hbar\delta_{jk}) \tag{21}$$

$$= \epsilon_{ijk} \hat{x}_j \hat{p}_k - i\hbar\epsilon_{ijk}\delta_{jk}. \tag{22}$$

The second term contains the contraction of the antisymmetric tensor ϵ_{ijk} with the symmetric Kronecker delta δ_{jk}. Since ϵ_{ijk} vanishes if any two indices are equal (which δ_{jk} forces), we have:

$$\epsilon_{ijk}\delta_{jk} = \epsilon_{ijj} = 0. \tag{23}$$

Therefore:

$$\hat{L}_i^\dagger = \epsilon_{ijk}\hat{x}_j\hat{p}_k = \hat{L}_i. \tag{24}$$

Conclusion: The operator $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ is inherently Hermitian. The specific non-commutativity of $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ does not affect the hermiticity of $\hat{\mathbf{L}}$ because the cross product couples orthogonal components (e.g., x and p_y) which mutually commute ($[\hat{x}, \hat{p}_y] = 0$).

3.3 Explicit Component Form

Expanding the tensor notation ($\hat{L}_x$ corresponds to $i = 1$, etc.), we obtain the explicit forms used in standard calculations:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \tag{25}$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \tag{26}$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x. \tag{27}$$

These expressions confirm that the quantum angular momentum operator preserves the functional form of its classical counterpart, provided the order of operators is maintained as $\mathbf{r} \times \mathbf{p}$.

3.4 Geometric Algebra Formulation: Derivation from the Fundamental Product

A more profound insight into the nature of angular momentum can be obtained using Geometric Algebra (Clifford Algebra $\mathcal{C}\updownarrow_{3,0}$). In this framework, the fundamental operation is the **Geometric Product**, denoted by the juxtaposition of vectors $\hat{\mathbf{r}}\hat{\mathbf{p}}$.

For any two vectors, the geometric product decomposes into a symmetric scalar part (Grade 0) and an antisymmetric bivector part (Grade 2):

$$\hat{\mathbf{r}}\hat{\mathbf{p}} = \hat{\mathbf{r}} \cdot \hat{\mathbf{p}} + \hat{\mathbf{r}} \wedge \hat{\mathbf{p}}. \tag{28}$$

Here, $\hat{\mathbf{r}} \cdot \hat{\mathbf{p}}$ is the standard scalar product (related to dilation), and $\hat{\mathbf{r}} \wedge \hat{\mathbf{p}}$ is the **exterior product** (wedge product).

3.4.1 Isolating the Bivector of Rotation

The angular momentum is physically defined as the generator of rotations. In Geometric Algebra, rotations are generated by bivectors. To isolate this physical quantity from the fundamental geometric product, we employ the **grade projection operator** $\langle \ldots \rangle_k$:

$$\hat{\mathcal{L}} \equiv \langle \hat{\mathbf{r}}\hat{\mathbf{p}} \rangle_2 = \hat{\mathbf{r}} \wedge \hat{\mathbf{p}}. \tag{29}$$

Unlike the vector cross product, which is limited to three dimensions, the bivector $\hat{\mathcal{L}}$ represents the *oriented plane* of rotation and is valid in any dimension.

3.4.2 Relation to the Cross Product (Hodge Dual)

The assignment requires the definition $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$. This vector $\hat{\mathbf{L}}$ is merely the **Hodge dual** of the fundamental bivector $\hat{\mathcal{L}}$. In 3D Euclidean space, the unit pseudoscalar $I = \hat{e}_1\hat{e}_2\hat{e}_3$ maps bivectors to vectors via the duality relation:

$$\hat{\mathbf{L}} = -iI(\hat{\mathbf{r}} \wedge \hat{\mathbf{p}}). \tag{30}$$

Thus, the standard operator $\hat{\mathbf{L}}$ is the vector normal to the bivector plane $\hat{\mathcal{L}}$. While both definitions are mathematically consistent in 3D, the geometric product formulation $\hat{\mathbf{r}}\hat{\mathbf{p}}$ reveals that angular momentum arises naturally as the antisymmetric component of the interaction between position and momentum.

4 Representation of $\hat{L}$ in Position Space

In the position representation, the Hilbert space is spanned by the eigenkets $|\mathbf{r}\rangle$. The position operator acts multiplicatively, $\hat{\mathbf{r}} \to \mathbf{r}$, and the linear momentum operator acts as the gradient generator.

4.1 Representation in Cartesian Coordinates

4.1.1 Derivation from Fundamental Operators

Using the position representation of the momentum operator[cite: 42]:

$$\hat{\mathbf{p}} = -i\hbar\nabla_{\mathbf{r}}, \tag{31}$$

we substitute this into the definition of angular momentum derived in the previous section ($\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$):

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times (-i\hbar\nabla) = -i\hbar(\mathbf{r} \times \nabla). \tag{32}$$

4.1.2 Explicit Components

Expanding the cross product in Cartesian coordinates using the determinant mnemonic (or the Levi-Civita definition):

$$\mathbf{r} \times \nabla = \begin{vmatrix} \hat{\mathbf{e}}_x & \hat{\mathbf{e}}_y & \hat{\mathbf{e}}_z \\ x & y & z \\ \partial_x & \partial_y & \partial_z \end{vmatrix} = \hat{\mathbf{e}}_x(y\partial_z - z\partial_y) - \hat{\mathbf{e}}_y(x\partial_z - z\partial_x) + \hat{\mathbf{e}}_z(x\partial_y - y\partial_x). \tag{33}$$

Multiplying by the factor $-i\hbar$, we obtain the explicit components[cite: 45]:

$$\hat{L}_x = -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \tag{34}$$

$$\hat{L}_y = -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \tag{35}$$

$$\hat{L}_z = -i\hbar \left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \right). \tag{36}$$

4.2 Representation in Spherical Coordinates

We now transform these operators to spherical coordinates (r, θ, ϕ) defined by:

$$\begin{aligned} x &= r\sin\theta\cos\phi, \\ y &= r\sin\theta\sin\phi, \\ z &= r\cos\theta. \end{aligned}$$

4.2.1 Derivation of $\hat{L}_z$

To find the expression for $\hat{L}_z$ in spherical coordinates, we apply the Chain Rule to the partial derivative with respect to ϕ. Let $\psi(x, y, z)$ be a function of position. The derivative with respect to ϕ is:

$$\frac{\partial}{\partial \phi} = \frac{\partial x}{\partial \phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi}\frac{\partial}{\partial z}. \tag{37}$$

Calculating the partial derivatives of the coordinates:

- $\frac{\partial x}{\partial \phi} = \frac{\partial}{\partial \phi}(r\sin\theta\cos\phi) = -r\sin\theta\sin\phi = -y.$
- $\frac{\partial y}{\partial \phi} = \frac{\partial}{\partial \phi}(r\sin\theta\sin\phi) = r\sin\theta\cos\phi = x.$
- $\frac{\partial z}{\partial \phi} = 0$ (since z does not depend on ϕ).

Substituting these back into the chain rule expression:

$$\frac{\partial}{\partial \phi} = (-y)\frac{\partial}{\partial x} + (x)\frac{\partial}{\partial y} = x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}. \tag{38}$$

Comparing this result with Eq. (36), we immediately identify the term in parentheses. Thus:

$$\hat{L}_z = -i\hbar \left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \right) = -i\hbar\frac{\partial}{\partial \phi}. \tag{39}$$

4.2.2 Derivation of $\hat{L}^2$

To derive the squared total angular momentum $\hat{L}^2 = \hat{\mathbf{L}} \cdot \hat{\mathbf{L}}$, we avoid the tedious squaring of Cartesian components and instead use the vector properties of the Gradient in spherical coordinates.

Step 1: The Gradient in Spherical Coordinates The gradient operator ∇ is expressed in the spherical orthonormal basis $(\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi)$ as:

$$\nabla = \hat{\mathbf{e}}_r\frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta\frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}. \tag{40}$$

Step 2: The Vector $\hat{\mathbf{L}}$ in Spherical Basis Using $\mathbf{r} = r\hat{\mathbf{e}}_r$ and calculating the cross product $\hat{\mathbf{L}} = -i\hbar(\mathbf{r} \times \nabla)$:

$$\hat{\mathbf{L}} = -i\hbar \left[r\hat{\mathbf{e}}_r \times \left(\hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r \sin\theta} \frac{\partial}{\partial \phi} \right) \right]. \tag{41}$$

Using the orthogonality relations $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_r = 0$, $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\phi$, and $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi = -\hat{\mathbf{e}}_\theta$:

$$\begin{aligned} \hat{\mathbf{L}} &= -i\hbar \left[\left(r\frac{1}{r} \right) (\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta) \frac{\partial}{\partial \theta} + \left(r \frac{1}{r \sin\theta} \right) (\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\phi) \frac{\partial}{\partial \phi} \right] \\ &= -i\hbar \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \hat{\mathbf{e}}_\theta \frac{1}{\sin\theta} \frac{\partial}{\partial \phi} \right). \end{aligned} \tag{42}$$

Step 3: Calculating $\hat{L}^2 = \hat{\mathbf{L}} \cdot \hat{\mathbf{L}}$ We apply the vector operator $\hat{\mathbf{L}}$ from Eq. (42) to itself. Extreme care must be taken because the unit vectors $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\phi$ depend on the coordinates, so derivatives acting on them are non-zero.

$$\hat{L}^2 = (-i\hbar)^2 \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right) \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} - \frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right). \tag{43}$$

Expanding the dot product (remembering that the first operator acts on everything to its right, including unit vectors):

$$\frac{\hat{L}^2}{-\hbar^2} = \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta} \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} \right)}_{\text{Term 1}} - \underbrace{\hat{\mathbf{e}}_\phi \cdot \frac{\partial}{\partial \theta} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right)}_{\text{Term 2}} - \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi} \left(\hat{\mathbf{e}}_\phi \frac{\partial}{\partial \theta} \right)}_{\text{Term 3}} + \underbrace{\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial \phi} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \frac{\partial}{\partial \phi} \right)}_{\text{Term 4}}. \tag{44}$$

Required Derivatives of Unit Vectors:

- $\partial_\theta \hat{\mathbf{e}}_\phi = 0$.
- $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$ (Orthogonal to $\hat{\mathbf{e}}_\phi$, so dot product vanishes).
- $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin\theta \hat{\mathbf{e}}_r - \cos\theta \hat{\mathbf{e}}_\theta$.
- $\partial_\phi \hat{\mathbf{e}}_\theta = \cos\theta \hat{\mathbf{e}}_\phi$.

Evaluating Terms:

Term 1: Since $\partial_\theta \hat{\mathbf{e}}_\phi = 0$:

$$\hat{\mathbf{e}}_\phi \cdot \left(\hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \theta^2} \right) = \frac{\partial^2}{\partial \theta^2}.$$

Term 2: Contains $\hat{\mathbf{e}}_\phi \cdot \partial_\theta \hat{\mathbf{e}}_\theta$. Since $\partial_\theta \hat{\mathbf{e}}_\theta = -\hat{\mathbf{e}}_r$, the dot product with $\hat{\mathbf{e}}_\phi$ is 0. Cross terms vanish here.

Term 3:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \left((\partial_\phi \hat{\mathbf{e}}_\phi) \frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{\partial^2}{\partial \phi \partial \theta} \right).$$

Using $\partial_\phi \hat{\mathbf{e}}_\phi = -\sin\theta \hat{\mathbf{e}}_r - \cos\theta \hat{\mathbf{e}}_\theta$:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \left(-\cos\theta \hat{\mathbf{e}}_\theta \frac{\partial}{\partial\theta} \right) = -\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\theta} = -\cot\theta \frac{\partial}{\partial\theta}.$$

Term 4:

$$\frac{\hat{\mathbf{e}}_\theta}{\sin\theta} \cdot \frac{\partial}{\partial\phi} \left(\frac{\hat{\mathbf{e}}_\theta}{\sin\theta}\frac{\partial}{\partial\phi} \right).$$

The derivative acts on $\hat{\mathbf{e}}_\theta$ and the function. However, $\partial_\phi \hat{\mathbf{e}}_\theta = \cos\theta \hat{\mathbf{e}}_\phi$, which is orthogonal to the pre-factor $\hat{\mathbf{e}}_\theta$. Thus, only the derivative on the scalar function survives:

$$\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}.$$

Assembly: Summing the non-vanishing contributions:

$$\frac{\hat{L}^2}{-\hbar^2} = \frac{\partial^2}{\partial\theta^2} + \cot\theta \frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}. \tag{45}$$

Recognizing that $\frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)$, we arrive at the final expression[cite: 48]:

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \right]. \tag{46}$$

5 Angular Momentum as the Generator of Rotations

In this section, we establish the fundamental role of the angular momentum operator $\hat{\mathbf{L}}$ as the generator of spatial rotations. We proceed from the definition of the unitary rotation operator to the conservation laws derived from the symmetry of the Hamiltonian.

5.1 Finite Rotation Operator

5.1.1 Definition

We define the rotation of a physical system by an angle $|\boldsymbol{\theta}|$ around an axis defined by the unit vector $\mathbf{n} = \boldsymbol{\theta}/|\boldsymbol{\theta}|$ using the exponential map. The unitary operator for a finite rotation is defined as[cite: 53]:

$$\hat{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{\hbar}\boldsymbol{\theta}\cdot\hat{\mathbf{L}} \right). \tag{47}$$

Since $\hat{\mathbf{L}}$ is Hermitian, $\hat{R}(\boldsymbol{\theta})$ is unitary ($\hat{R}^\dagger = \hat{R}^{-1}$), ensuring the conservation of probability (norm of the state) during rotation.

5.1.2 Action on Position Kets

To demonstrate the action of $\hat{R}(\boldsymbol{\theta})$ on the position eigenkets $|\mathbf{r}\rangle$, we consider the transformation of the position operator $\hat{\mathbf{r}}$. Using the Baker-Campbell-Hausdorff lemma or the infinitesimal properties derived below, one can show that:

$$\hat{R}^\dagger(\boldsymbol{\theta})\hat{\mathbf{r}}\hat{R}(\boldsymbol{\theta}) = \mathcal{R}(\boldsymbol{\theta})\hat{\mathbf{r}}, \tag{48}$$

where $\mathcal{R}(\boldsymbol{\theta})$ is the classical 3×3 rotation matrix. Applying this to a position eigenket $|\mathbf{r}\rangle$:

$$\hat{\mathbf{r}}\left(\hat{R}(\boldsymbol{\theta})|\mathbf{r}\rangle\right) = \hat{R}(\boldsymbol{\theta})\left(\hat{R}^\dagger\hat{\mathbf{r}}\hat{R}\right)|\mathbf{r}\rangle = \hat{R}(\boldsymbol{\theta})(\mathcal{R}\mathbf{r})|\mathbf{r}\rangle. \tag{49}$$

Since $|\mathbf{r}\rangle$ is an eigenstate of $\hat{\mathbf{r}}$ with eigenvalue $\mathbf{r}$, the rotated ket must be an eigenstate of the position operator with eigenvalue $\mathcal{R}\mathbf{r}$. Thus[cite: 55]:

$$\hat{R}(\boldsymbol{\theta})|\mathbf{r}\rangle = |\mathcal{R}(\boldsymbol{\theta})\mathbf{r}\rangle. \tag{50}$$

Physically, this implies that the operator actively transports a particle localized at $\mathbf{r}$ to the rotated point $\mathcal{R}\mathbf{r}$.

5.2 Infinitesimal Rotation

5.2.1 First-Order Expansion

Consider an infinitesimal rotation by an angle $d\boldsymbol{\theta}$. Expanding the exponential to first order in the parameter $d\boldsymbol{\theta}$[cite: 59]:

$$\hat{R}(d\boldsymbol{\theta}) \approx \hat{\mathbb{I}} - \frac{i}{\hbar}d\boldsymbol{\theta}\cdot\hat{\mathbf{L}}. \tag{51}$$

5.2.2 Action on the Wavefunction

We examine how this operator transforms the wavefunction $\psi(\mathbf{r}) = \langle\mathbf{r}|\psi\rangle$. The transformed state $|\psi'\rangle = \hat{R}(d\boldsymbol{\theta})|\psi\rangle$ has the wavefunction:

$$\psi'(\mathbf{r}) = \langle\mathbf{r}|\hat{R}(d\boldsymbol{\theta})|\psi\rangle \approx \langle\mathbf{r}|\left(1 - \frac{i}{\hbar}d\boldsymbol{\theta}\cdot\hat{\mathbf{L}}\right)|\psi\rangle. \tag{52}$$

Distributing the bra-ket:

$$(\hat{R}(d\boldsymbol{\theta})\psi)(\mathbf{r}) \approx \psi(\mathbf{r}) - \frac{i}{\hbar}d\boldsymbol{\theta}\cdot\langle\mathbf{r}|\hat{\mathbf{L}}|\psi\rangle. \tag{53}$$

Using the position representation of angular momentum derived in Section 3, $(\hat{\mathbf{L}}\psi)(\mathbf{r}) = -i\hbar(\mathbf{r}\times\nabla)\psi(\mathbf{r})$, we substitute:

$$(\hat{R}(d\boldsymbol{\theta})\psi)(\mathbf{r}) \approx \psi(\mathbf{r}) - \frac{i}{\hbar}d\boldsymbol{\theta}\cdot\left[-i\hbar(\mathbf{r}\times\nabla)\psi(\mathbf{r})\right]. \tag{54}$$

$$(\hat{R}(d\boldsymbol{\theta})\psi)(\mathbf{r}) \approx \psi(\mathbf{r}) - d\boldsymbol{\theta} \cdot (\mathbf{r} \times \nabla)\psi(\mathbf{r}). \tag{55}$$

Using the vector identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$, we rewrite the correction term:

$$d\boldsymbol{\theta} \cdot (\mathbf{r} \times \nabla) = (d\boldsymbol{\theta} \times \mathbf{r}) \cdot \nabla. \tag{56}$$

Recognizing that $d\mathbf{r} = d\boldsymbol{\theta} \times \mathbf{r}$ is the displacement vector due to rotation, we obtain the Taylor expansion for a scalar field under infinitesimal coordinate displacement[cite: 61]:

$$\psi'(\mathbf{r}) \approx \psi(\mathbf{r}) - d\mathbf{r} \cdot \nabla\psi(\mathbf{r}) \approx \psi(\mathbf{r} - d\mathbf{r}). \tag{57}$$

This confirms consistency with the passive transformation view: rotating the state active forward is equivalent to evaluating the original function at the reverse-rotated coordinates.

5.3 Symmetries and Conservation Laws

5.3.1 Commutation with the Free Hamiltonian

The free particle Hamiltonian is $\hat{H} = \frac{\hat{p}^2}{2m}$. To test for rotational invariance, we compute the commutator $[\hat{H}, \hat{L}_i]$[cite: 64]. Since $\hat{p}^2$ is a scalar operator (dot product of a vector operator with itself), it should commute with the generator of rotations. Explicitly:

$$[\hat{L}_i, \hat{p}^2] = \sum_j [\hat{L}_i, \hat{p}_j\hat{p}_j] = \sum_j \left(\hat{p}_j[\hat{L}_i, \hat{p}_j] + [\hat{L}_i, \hat{p}_j]\hat{p}_j\right). \tag{58}$$

Using the canonical commutation relation for angular momentum and vector operators, $[\hat{L}_i, \hat{p}_j] = i\hbar\epsilon_{ijk}\hat{p}_k$:

$$[\hat{L}_i, \hat{p}^2] = i\hbar\sum_{j,k} \epsilon_{ijk}(\hat{p}_j\hat{p}_k + \hat{p}_k\hat{p}_j) = 2i\hbar\sum_{j,k} \epsilon_{ijk}\hat{p}_j\hat{p}_k. \tag{59}$$

This sum involves the contraction of the antisymmetric tensor ϵ_{ijk} (in indices j, k) with the symmetric tensor $\hat{p}_j\hat{p}_k$. Therefore, the sum vanishes identically:

$$[\hat{H}, \hat{\mathbf{L}}] = 0. \tag{60}$$

5.3.2 Conservation of Angular Momentum

We invoke Ehrenfest's Theorem, established in Section 1[cite: 32]:

$$\frac{d}{dt}\langle\hat{\mathbf{L}}\rangle = \frac{1}{i\hbar}\langle[\hat{\mathbf{L}}, \hat{H}]\rangle. \tag{61}$$

Substituting the zero commutator derived above:

$$\frac{d}{dt}\langle\hat{\mathbf{L}}\rangle = 0. \tag{62}$$

This result [cite: 66] constitutes the quantum mechanical proof of the conservation of angular momentum for a free particle.

5.3.3 Connection to the Correspondence Principle

Classically, for a free particle, there is no torque ($\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = 0$), implying $d\mathbf{L}_{cl}/dt = 0$. The quantum result perfectly mirrors the classical law, as guaranteed by the Correspondence Principle. The rotational symmetry of space (isotropy) leads to the conservation of angular momentum in both frameworks, with the operator $\hat{\mathbf{L}}$ playing the dual role of the observable quantity and the generator of the symmetry group $SO(3)$.